RanPaste: Paste Consistency and Pseudo Label for Semisupervised Remote Sensing Image Semantic Segmentation

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***Abstract*— With the development of deep learning, remote sensing (RS) image segmentation has been applied with marked success. However, in the process of model training, the large number of labeled images required more expensive annotation. A key challenge is how to make full use of extensive unlabeled images available to improve the segmentation model. In this article, we propose a semisupervised remote sensing image semantic segmentation method defined as RanPaste, which combines labeled images with unlabeled images to improve segmentation performance. First, we obtain pseudo label by randomly pasting part of the ground truth label into the predicted segmentation map. Then, we combine the labeled and unlabeled images to generate rough predictions after strong augmentation. Finally, by using the semisupervised loss, we achieve better performance on remote sensing image segmentation. Our method combines consistency regularization and pseudo label and then utilizes thresholds to gradually improve the model performance. RanPaste enables the model to learn more underlying information in the unlabeled data. Experimental results on six datasets show that RanPaste can learn more latent information from unlabeled data to improve segmentation performance. Besides, our approach achieves better segmentation results on different network structures and datasets.**

***Index Terms*— Convolutional neural network, remote sensing (RS), semantic segmentation, semisupervised learning.**

I. Introduction

With the development of advanced aerial image sensing technology, remotely sensed data has become abundant. Scene classification and object detection of remote sensing (RS) images have been widely studied and applied in the past decade and have achieved good performance. In the processing of these remote sensing images, image semantic segmentation can obtain important ground information in remote sensing data. remote sensing image segmentation has many important applications in change detection [1], [2], road extraction [3], [4], loss estimation on pest and disease, forecast of agricultural harvest, and so on.

In the early research, most remote sensing image segmentation approaches usually used some traditional methods, such as superpixel segmentation, watershed segmentation, and level set segmentation. These methods do not require a lot of labeled data for training, but the final segmentation results are usually rough. With the successful application of deep learning (DL) in nature image processing, the deep network has gained more attention in remote sensing image. In the task of road extraction of remote sensing data, many researchers try to use different network models to segment remote sensing images. The good performance of these methods proves that DL has great application potential on remote sensing dataset. However, while DL-based methods achieve more accurate segmentation performance, these methods also have some limitations. Such methods require a large amount of annotated data that are often costly to obtain. Due to the high resolution and multispectral characteristics of remote sensing images, data annotation is usually more complex than nature images. Therefore, it is worth studying how to make full use of a large amount of unlabeled data to improve the model performance when only few labeled data are available.

In the DL method of nature images, semisupervised learning is an important method to solve the lack of annotated data. For image classification task, many methods [5], [6], [7] have made great progress on CIFAR 10 [8] and other datasets. These methods greatly improve the classification accuracy when only few annotated data are available. However, semantic segmentation aims to assign labels to each pixel in an image. In general, the semisupervised image classification method cannot be directly used in image segmentation, especially in remote sensing images. For the problem of insufficient available labeled data in image segmentation, some methods try to use image-level label or box label instead of pixel-level label. These weakly supervised methods have also achieved better performance. In recent years, some methods have tried to improve semisupervised image segmentation by using some perturbation methods. To leverage the unlabeled examples, Ouali *et al.* [9] enforced consistency between the main decode predictions and the auxiliary decoders and conducted an

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exhaustive study of perturbations for unlabeled images. They proposed a cross-consistency training (CCT) method for semisupervised semantic segmentation. Inspired by their approach, our method also learns some ways from their method of reducing overfitting.

Recently, we proposed a semisupervised remote sensing image segmentation method [10], which contains consistency regularization (CR) method and average update pseudo label (AUP). We verified that this method can effectively utilize unlabeled data on different remote sensing datasets. However, this method also has some problems. For the perturbation of unlabeled data, only color jitter and random noise are used, and the utilization of pseudo label is complex and unstable. Recent semisupervised approaches [11], [12] mix unlabeled images and achieve state-of-the-art performance for image classification. However, their method of mixing images directly that uses in segmentation will change labels. Given these problems, we propose a semisupervised remote sensing image segmentation method called RanPaste, which paste labeled images into unlabeled images for CR. Inspired by fixmatch [12], we paste part of the labeled image into the unlabeled image and mix them. Our method first paste part of the labeled image into the unlabeled image and to mix the ground truth and pseudo label. Our pasting method is to combine different image areas. In addition, for the semisupervised segmentation algorithm, we propose an adaptive threshold filter method for different pixels to improve the quality of pseudo labels. Finally, we use standard cross-entropy loss for both labeled and unlabeled data. In this way, we simplify the CR method and pseudo-label update algorithm and combine the output threshold to gradually improve the performance of the model during the training process. Our semisupervised method performs better on three remote sensing datasets than CR and AUP methods, and we verify the effectiveness of our method on more types of remote sensing data.

The main contributions of this article are as follows.

1) We propose a semisupervised remote sensing image segmentation method, which can significantly improve the performance of the model when only a few labeled data can be used.
2) Our method randomly pastes part of the labeled image into the unlabeled image by using the RanPaste method as a strong perturbation and then generates pseudo labels for weakly enhanced data.
3) We propose an adaptive threshold method to weight supervised loss and semisupervised loss, and they are used to reduce the overfitting problem and improve the quality of pseudo labels.
4) We validate the effectiveness of our semisupervised approach on six different types of remote sensing datasets, including building segmentation, road extraction, multiclass segmentation, unmanned aerial vehicle (UAV) aerial data, and cloud segmentation.

Scenes of remote sensing images are usually complex. Different from natural images, remote sensing images have high intraclass variance, while the interclass variance is low. Our random pasting method can improve the extraction of object edge information. By the combination of labeled images and unlabeled images, the semisupervised method can effectively improve this problem. Besides, different from general image segmentation tasks, it is more difficult to annotate the pixel-level labels of remote sensing images. However, it is easier to obtain the remote sensing data without labels. Therefore, considering these situations, our semisupervised segmentation method has great application potential.

In the following, we introduce RanPaste for semisupervised image segmentation. Section II details relevant remote sensing segmentation methods and related semisupervised approaches for remote sensing data. In Section III, we describe how our method can be applied to remote sensing images and our improvements for semisupervised image segmentation. Section IV shows the experimental results of RanPaste on five datasets. Then, in Section V, we discuss the ablation study and time complexity for our method. Finally, we conclude in Section VI with a summary and an outlook for future work.

II. Related Work

In this section, we first review the segmentation methods for remote sensing images. These methods include traditional segmentation approaches and some new segmentation methods based on DL. Then, we review some related semisupervised methods for remote sensing. Finally, we will introduce our new approach and compare it with some relevant effective methods.

A. Remote Sensing Image Semantic Segmentation

Remote sensing image segmentation has always been an important tool in various applications, including change detection, road extraction. The early methods were mainly divided into three categories: 1) superpixel segmentation; 2) watershed algorithms; and 3) level set segmentation. The superpixel algorithm is an important method in image processing, and it has been widely used in remote sensing image segmentation. Lang *et al*. [13] proposed a super pixel segmentation method that is combined with the generalized mean shift (GMS) algorithm. They introduce a presorting strategy into the GMS segmentation algorithm to improve the segmentation performance. In addition, the notion of the watershed edge-weighted graph was been used for watershed segmentation. For the segmentation of remote sensing images, Ciecholewski *et al*. [14] applied the watershed segmentation combined regions by maximizing the average contrast. This method achieved good performance on the synthetic aperture radar (SAR) datasets. The level set segmentation algorithm [15] combines an advanced statistical model and a powerful segmentation method. This approach obtains a better segmentation accuracy for the high-resolution PolSAR image segmentation problem.

Recently, DL image segmentation methods have also demonstrated improved performance. Long *et al*. [16] proposed fully convolutional network (FCN), which can accept any size of the input image, and the segmentation performance is greatly improved than conventional methods. Most of the later works on semantic segmentation are based on FCN. UNet [17] uses an encode–decode architecture to improve the segmentation results for medical image. This approach can also achieve good segmentation performance when fewer annotated training data are available. To solve the problem of increasing

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the receptive field of the features in semantic segmentation without decreasing the size of the feature map, DeepLab V1 [18], V3 [19], and V3+ [20] use dilated convolution to extract features from input images. In our experiments, we use UNet, DeepLab V3, and DeepLab V3+ as our basic network architectures, which are widely used in remote sensing image segmentation.

For the semantic segmentation of remote sensing images, there are advanced methods based on DL. These methods usually obtain more accurate segmentation results compared with the traditional segmentation approach. During the last two years, remote sensing image segmentation has been further studied in different applications, including building detection, road extraction, and cloud segmentation. Zhang *et al*. [3] proposed a DL network structure based on UNet for road extraction. This network combines the advantages of UNet and ResNet and has a good performance for road extraction on aerial images. For the road extraction task, Lin *et al*. [4] significantly improved by using different deep network architectures. For cloud detection, Jiao *et al*. [21] have proposed a DL segmentation method. Mou *et al*. [22] have introduced two effective network units, the spatial relation module and the channel relation module to produce relation-augmented (RA) feature representations. Zhou *et al*. [23] proposed a novel CGFDN for airborne image segmentation. This method utilizes the advantage of the co-occurrence relations between different classes of objects to improve the segmentation performance. Ding *et al*. [24] proposed both a patch attention module (PAM) to embed scene information from local patches and an attention embedding module (AEM) to enhance the semantic representation of low-level features by introducing attention from high-level features. Their local attention network (LANet) greatly improved the performance on RSI datasets. In our semisupervised experiments, we also use LANet as our baseline to compare our RanPaste with some networks specifically that are designed for remote sensing image segmentation.

In recent years, there are some new methods for remote sensing image classification. To improve the problem for spectral unmixing to accurately estimate abundance maps, Hong *et al*. [25] proposed a novel spectral mixture model called an augmented linear mixing model (ALMM). Besides, for hyperspectral image (HSI) classification, Hong *et al*. [26] analyzed CNNs and graph convolutional networks (GCNs) with a focus on HS image classification and proposed a mini GCNs for remote sensing data. Lately, some researchers propose a unified multimodel DL framework with a focus on the RS image classification [27], and this method can be generalized to cross-modality learning (CML)'s with more effective and compact modality blending. In our experiment on semisupervised remote sensing image segmentation, we use UNet, DeepLab V3, DeepLab V3+, and LANet networks as our baselines.

B. Semisupervised Learning on Remote Sensing Image

Although DL has achieved impressive results in image classification, image segmentation, and other tasks, it usually requires a large amount of annotated training data. Given these reasons, researchers are increasingly focusing on methods of semisupervised learning. For the semisupervised image classification, both ladder networks [28] and the temporal method [5] demonstrate that the idea of consistency loss can take full advantage of unlabeled data. Based on the consistency method, a mean teacher [6] updates the parameters of the model instead of the updated features. This method gives higher efficiency and the final classification accuracy is also improved. Based on the perturbation method for unlabeled data, the virtual adversarial training (VAT) method [7] is an algorithm that generates adversarial perturbations. By adding this noise to consistency, training can improve the robustness of the model and achieve better performance in semisupervised classification. More recently, FixMatch [12] combines the benefits of the effective approaches and makes better pseudo labels available during the training process. Inspired by the method of pseudo labels, our RanPaste method uses a similar method for remote sensing image segmentation.

Remote sensing datasets usually have the characteristics of high resolution and multispectral coverage, so the cost of annotation for the purpose of providing training data becomes prohibitively expensive. In comparison, large amounts of unlabeled remote sensing data are relatively easily obtainable. Recently, more methods have been improved the performance of models by using unlabeled remote sensing data. Zhang *et al*. [29] used a coarse-to-fine strategy to deal with the testing samples, to develop a more effective semisupervised change detection method. For the utilization of unlabeled data, there are also some studies that use unsupervised methods. An unsupervised deep change vector analysis method for very high spatial resolution (VHR) was proposed by Tasar *et al*. [30]. Their approach exploits suboptimal deep features extracted from pretrained multilayer CNN networks. On the task of semantic segmentation of remote sensing images, Wang *et al*. [31] used sparse pixel labels to perform weakly supervised image segmentation.

With the successful application of semisupervised learning in image classification, some semisupervised methods applied to remote sensing images have been proposed. Hong *et al*. [32] proposed a novel iterative multitask regression (IMR) framework with the application to semisupervised HDR. They make pseudo labels to be gradually updated using the learned graph in each outer iteration. Our pseudo labels are also designed to be updated once for each training batch so that more accurate pseudo labels can be obtained. X-ModalNet [33] is progressively optimizing pseudo labels to further find a better decision boundary. Sha *et al*. [34] proposed a method using edge-conditioned convolution for GCNs. They combined the spatial–spectral features of HSIs into the graph representation for remote sensing classification. Different from multimodal deep neural networks (DNNs), Hong *et al*. [35] proposed two novel plug-and-play units in the network to learn perturbation-insensitive feature representations and to eliminate the gap between multimodalities. Their self-GANs and mutual-GANs proved the effectiveness of semisupervised remote sensing image semantic segmentation using GAN-based methods.

Recently, we proposed the CR and AUP [10] method for semisupervised segmentation of remote sensing images. In this

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method, we use the consistency method in semisupervised learning and exploit the AUPs to guide the training process although this method performs well in the semisupervised segmentation of remote sensing images. ClassMix [36] improves the robustness of the model by mixing the input images; however, the class-mix method cannot be easily adapted to the characteristics of remote sensing images. The objects in remote sensing images are usually small, and the trained input images are both high resolution and multispectral. As a result, in our RanPaste method, we adopt the mixing method to better combine both labeled data and unlabeled data. Finally, we also use some elements of Fixmatch [12] and cross view [9] to prevent model overfitting. RanPaste combines the advantages of semisupervised learning, simplifies the training process, and improves the segmentation for remote sensing images.

III. Proposed Approach

We propose a semisupervised semantic segmentation method for remotely sensed images. Our method improves on the CR in semisupervised learning. Mixup [37] is an effective data augmentation method for the task of image classification. This method is based on the weighted sum of each pixel in different input images, and their pixel-level labels are mixed in the same way. Inspired by this method, we propose an image mixing method for semisupervised semantic segmentation. Unlike the Mixup method, we paste part of the labeled image into the unlabeled image, rather than directly mixing them. Since the task of semantic segmentation is to classify all pixels, we change the values of some pixels and their corresponding labels at the same time. For the randomly selected paste area M, we use M to paste part of the ground truth into the predicted result of the unlabeled image. Our approach uses this randomly paste method as a data perturbation for semisupervised learning.

Because of these improvements for the image mixing method, our new sequential method is more suitable for semisupervised learning and segmentation tasks. By using labeled images paste into the unlabeled images, supervised learning can be better utilized to improve semisupervised training in the training process. Next, we will introduce more details about our semisupervised approach, including the idea of the RanPaste and our improvements of the pseudo-label algorithm.

A. RanPaste

For semisupervised semantic segmentation of remote sensing images, let $X = \{(x_l, y_l) : l \in (1, \ldots, L)\}$ represent a dataset with labeled remote sensing images. x_l are the labeled images, y_l are the corresponding pixel-level labels, and L is the number of labeled training examples. Let $U = \{(u_{xb}) : b \in (1, \ldots, B)\}$ be the dataset of unlabeled examples, where u_{xb} are unlabeled images and B is the number of unlabeled images. We use the mean teacher method as our baseline in the model training process. Then, parameters of the teacher model are θ_t, and the parameters of the student model are θ_s. In the training process, only the parameters of the student model participate in the backpropagation, and the parameters of the teacher model smoothly update according to the latest student model.

In the general semisupervised method, semisupervised loss only contains unlabeled data, but in our RanPaste method, we combine the labeled data with unlabeled data by pasting images. We believe that the most important thing is that we combine pseudo label and ground truth, so we strengthen the ability of the model to extract edge information. Besides, by mixing and pasting images, the semisupervised model will gradually increase the difference of decision boundaries.

The main idea of our RanPaste method is to design a semisupervised method suitable for semantic segmentation on remotely sensed images. For the CR to become commonly used in semisupervised learning, we use the random paste as a perturbation method. Then, the random noise and color jitter are combined and added to further strengthen the input images for perturbation so that the output result of the student model is more robust. At the same time, for the original images, we conduct weak image augmentation and then, through a more accurate teacher model, obtain a more refined segmentation result. For these two kinds of segmentation results, the general semisupervised method is to use consistency loss. In our method, we adopt instead standard cross-entropy loss and the pseudo-label method to optimize the model simply and efficiently. Next, we will introduce the training process for the semisupervised method RanPaste.

For annotated remote sensing images X, we use the standard supervised learning method to train the student model. We add color jitter to the input image x and then add random noise. We input the enhanced images to obtain the predicted results of the model. Our approach uses the general cross-entropy loss [36] as the loss in the supervised learning

$$H(y, f(x)) = -\frac{1}{W \cdot H} \sum_{i=1}^{W} \sum_{j=1}^{H} \sum_{c=1}^{C} y_{i,j,c} \cdot \log f(x)_{i,j,c} \quad (1)$$

where $y_{i,j,c}$ is the label that the pixel in coordinates i and j belongs to class c according to the ground truth. Let $f(x)_{i,j,c}$ be the predicted probabilities produced by the model for each pixel in the input image x. W and H are the width and height of the remote images, respectively, and C is the number of classes to be identified. The train of our segmentation model is to use cross-entropy loss for each pixel to optimize their prediction results closer to the ground truth.

Meanwhile, in order to improve the generalization ability of the model, our supervised loss is

$$L_r = \frac{1}{W \cdot H} \sum_{i=1}^{W} \sum_{j=1}^{H} \sum_{c=1}^{C} f(x)_{i,j,c} \log f(x)_{i,j,c} \quad (2)$$

$$L_s = \frac{1}{|X|} \sum_{x \in X} H(y_x, f(x, \theta_s, \eta_w)) + \frac{\lambda}{|X|} \sum_{x \in X} L_r \quad (3)$$

where y_x is the label for input image x, θ_s is the student model parameters, and η_w is the weakly perturbation for input images. To prevent the model from overfitting few labeled data, we also add a regularization item L_r for supervised loss and λ is the regularization coefficient. This regularization is used to penalize the complexity of the label model.

As shown in Fig. 1, for the unlabeled remote sensing image u_x from the dataset U, we randomly select a pasted

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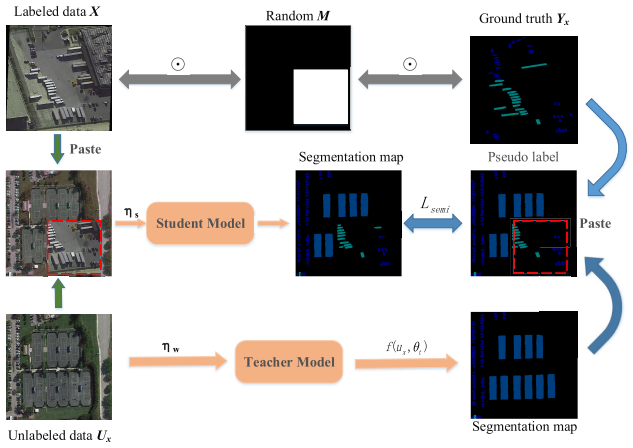


Fig. 1. Overview of the proposed semisupervised RanPaste method. Labeled images and labels are combined with mask M, respectively, to obtain the mixing inputs and pseudo labels. Finally, the semisupervised loss is calculated by combining the strong augmentation outputs with new pseudo labels.

region as M. We paste the corresponding area of the labeled image into the same area of the unlabeled image. By this random pasting method, we replace some pixels in the unlabeled image with pixels in the labeled image. Our method generates new boundaries for the objects in the remote sensing image, which is conductive to improving the robustness of the model in the training process. The input image after mixing is u'_x

$$u'_x = x \odot M + u_x \odot (1 - M). \tag{4}$$

M is a randomly selected paste area, which is a binary mask with the same size as the original image. The circ dot operator $\odot$ is the corresponding position multiplication of the original image matrix and the matrix of the binary mask M. By multiplying labeled and unlabeled images with their masks and adding them together, we mixed them to obtain u'_x.

For the input image u'_x, we add color jitter to it and then add a random strong Gaussian noise. A strong-augmented version of an unlabeled image is fed into the student model to obtain its predictions $f(u'_x, \theta_s, \eta_s)$. At the same time, we fed the same unlabeled image into the teacher network after weak data augmentation. The intensity of data augmentation is controlled by the coefficients of the color jitter and random noise. In the pseudo label mixing process, we have the teacher model prediction $f(u_x, \theta_t, \eta_w)$ and the true label y_x for labeled data x, where θ_t are teacher model parameters. As can be seen from Fig. 1, since image segmentation needs to keep the labels corresponding to each pixel consistent, we paste label of the labeled image into the predicted map. Then, we mix them to generate pseudo labels as follows:

$$y_p = y_x \odot M + f(u_x, \theta_t, \eta_w) \odot (1 - M). \tag{5}$$

Finally, for unlabeled data u_x, we obtain the student model output and weakly augmentation pseudo labels y_p. The semisupervised loss also uses cross-entropy loss, but we replace the normal label with the generated pseudo label

$$L_{\text{semi}} = \frac{1}{|U_x|} \sum_{u_x \in U_x} H(y_p, f(u'_x, \theta_s, \eta_s)). \tag{6}$$

Therefore, in the training process of our approach, we conduct both supervised training and unsupervised training. The final loss is given as follows:

$$L_{\text{all}} = L_s + L_{\text{semi}} \tag{7}$$

where L_s is the supervised loss and L_{semi} is the semisupervised loss, and they are applied for optimization. From (7), we can learn that we do not set training weights for supervised loss and semisupervised loss. Therefore, although L_s and L_{semi} have the same weight coefficient, our thresholds t_x and t_u filter them, respectively. In the training process, the weight of pseudo-label loss gradually increases from small to large. The specific reasons will be introduced in Section III-B.

The method we use the CR in semisupervised is to paste part of labeled images into unlabeled images. The RanPaste method is similar to random perturbations in general semisupervised learning methods. Since the same RanPaste method is also adopted for labels, the loss training of the original pixels in the unlabeled image will not be affected in the process of training. In general, our RanPaste method is a more appropriate CR method for remote sensing image segmentation.

B. Overfitting and Pseudo Label

In general semisupervised learning approaches, there are only few labeled remotely sensed images available, usually accounting for 1/8 or 1/4 of the total dataset. Therefore, in many semisupervised methods, model overfitting of labeled data is a common problem. In some studies, the method to solve this problem is that each batch contains only a small amount of labeled data and more unlabeled images to slow down the fitting speed of the model to labeled data. Some other methods add extra weights for supervised loss and semisupervised loss to balance the fitting degree of the model for different data.

In order to alleviate the problem of overfitting for labeled data in the training process, we conduct extra process for the supervised loss. The main purpose of our threshold t_x is to remove pixels with high predicted probabilities so that they do not participate in backpropagation. By removing these pixels, we reduce the overfitting problem of the model. For pixels with low predicted probabilities, we keep their loss the same as full supervised segmentation. These complex remote sensing objects can normally participate in the model optimization. For the labeled image x, we get a prediction map y'_x with the same size as the original image through the model. Each value in y_p is the highest probability that the pixel is predicted to be a class. According to the value of the predicted probability in each pixel, we obtain a loss mask α. The selective method of α is given as follows:

$$\alpha_{i,j} = \begin{cases} 1, & \text{if } \max\limits_{1 \le c \le C} (f(x)_{i,j,c}) < t_x \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

Let $f(x)_{i,j,c}$ be the predicted probability of the image x at the pixel (i, j) for class c. We use $\max\limits_{1 \le c \le C} (f(x)_{i,j,c})$ as the maximum predicted probability in each pixel. As shown in Fig. 2, when the predicted probability is greater than the threshold value t_x, the loss mask of this pixel is defined

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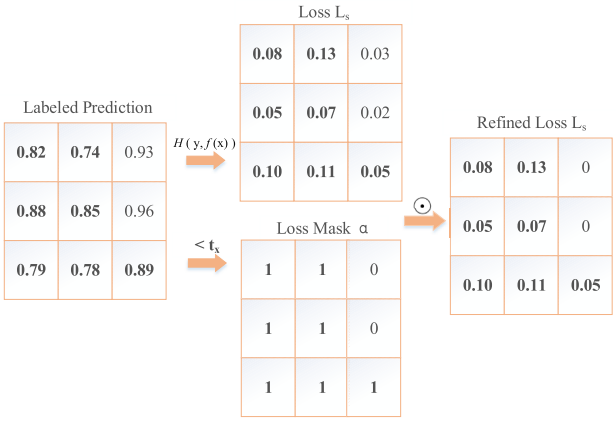


Fig. 2. Some examples of the calculation for supervised loss mask α when the threshold t_x is 0.9.

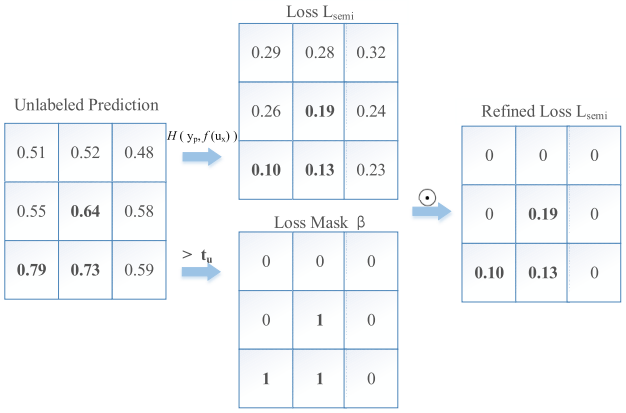


Fig. 3. Some examples of the calculation for semisupervised loss mask β when the threshold t_u is 0.6.

as 0, and the others are 1. Therefore, we filter the loss of labeled images. We select pixels with relatively low predicted probability to participate in backpropagation to optimize the model and ignore some pixels with high predicted probability that are easy to overfit the model.

With the training of the model, the accuracy of the prediction is gradually improved. Thus, we use a dynamically changing threshold t_x to calculate the loss mask α. t_x is defined as follows:

$$t_x = \frac{1}{W \cdot H} \sum_{i=1}^{W} \sum_{j=1}^{H} \text{sigmoid}\left(\max_{1 \le c \le C} (f(x)_{i,j,c}) \right) + 0.2. \quad (9)$$

We use the average predicted probability of all pixels in the image as the loss threshold. With the improvement of model accuracy, the probability of pixel prediction increases gradually, so the threshold t_x also increases through the training process. By adjusting the t_x value, we can properly control the fitting ability of the model in the training process. We calculate by sigmoid function to ensure the stability of the threshold value and add 0.2 to increase the threshold for labeled images. Through the dynamic adjustment of the threshold, the supervised training improves the performance of the model more stably.

In addition, the pseudo-label method is used in our semisupervised training, so we analyze the generation of pseudo labels in detail. Like most pseudo-label methods, we also assign labels to pixels whose predicted probability is greater than a threshold value. However, the difference is that we do not assign labels to the unlabeled data after the completion of all training processes. Instead, during the model training process, each batch of unlabeled data uses the current model to generate new pseudo labels.

As shown in Fig. 3, when the probability in the unlabeled predicted result is greater than t_u, the loss mask of this pixel is set to 1, and the other pixels are set to 0. Since the accuracy of the model will gradually increase in the training process, more pixels mask will be set to 1 for unlabeled images. In this way, we can also get a weight mask β corresponding to the unlabeled data

$$\beta_{i,j} = \begin{cases} 1, & \text{if } \max\limits_{1 \le c \le C} (f(u_x)_{i,j,c}) > t_u \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Let $f(u_x)_{i,j,c}$ be the predicted probability of the unlabeled image u_x at the pixel (i, j) for class c. We also use $\max_{1 \le c \le C}(f(u_x)_{i,j,c})$ as the maximum predicted probability in each pixel. Similar to the t_x, we use the average value of the model's predicted probability for pixels in the image as the current threshold for the model training. For the calculation threshold t_u of semisupervised loss mask β, we define t_u as

$$t_u = \frac{1}{W \cdot H} \sum_{i=1}^{W} \sum_{j=1}^{H} \text{sigmoid}\left(\max_{1 \le c \le C} (f(u_x)_{i,j,c}) \right). \quad (11)$$

As shown in Fig. 3, we multiply the corresponding pixels of loss and loss mask β to obtain the final semisupervised loss. With the gradual increase of threshold value, our semisupervised loss calculation requires greater prediction probability, so the model can learn more deep information from unlabeled data to improve the segmentation performance.

Using this improved pseudo-label method, our model continuously learns more information contained in the unlabeled remote sensing images while ensuring the stable improvement of the model performance in the training process. After combining the masks α and β of supervised and semisupervised loss, the final loss definition of our RanPaste method is given as follows:

$$L_{\text{all}} = \frac{1}{|X|} \sum_{x \in X} (\alpha \odot L_s) + \frac{1}{|U_x|} \sum_{u_x \in U_x} (\beta \odot L_{\text{semi}}). \quad (12)$$

We refine the supervised loss L_s and the semisupervised loss L_{semi} for input images. The final loss is a combination of (7) and our threshold methods. In the end, our total loss is still L_s and L_{semi} working together with the model. We add weights to them by the threshold methods and finally obtain the total loss L_{all} for optimization.

By setting the threshold t_x, our method will improve the model overfitting of few labeled data and meanwhile improve the robustness of the model. For the threshold t_u,

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TABLE I
COMPARISON OF FIVE REMOTE SENSING DATASETS

Dateset	Train / Test	Image Size	Data Source	Spectral Channels	Class
INRIA [38]	155 / 25	5000×5000	Inria Aeria	3	1
Road Extraction [39]	5400 / 826	1024×1024	Digital Globe	3	1
iSAID [40]	1200 / 211	455×387 - 4898×4242	Google Earth, GF-2, JL-1	3	15
AeroScapes [41]	2400 / 869	1280×720	UAV	3	11
38-Cloud [42]	4200 / 955	384×384	Landsat 8	4	1
Vaihingen	12 / 4	1996×1995 - 3816×2550	Airborne Sensors	3	5

the predicted probability of our model for unlabeled images will gradually increase so that more pixels will participate in the optimization process of the model. For predicted unstable or difficult to identify areas, the model ignores these areas during training. When the predicted accuracy increases to obtain more accurate segmentation maps, most objects can be added to the training process. Combining these two methods, our semisupervised training process can be steadily improved and the semisupervised model can gradually learn more information from unlabeled data. In the following part of the experiment, we have relative comparison experiments for two thresholds.

For labeled images, since there are few trainable labeled images, we set the threshold t_x to prevent the model from overfitting them. t_x is mainly used to dropout those pixels with high predicted probability and to improve the model by setting the selected loss to 0. For unlabeled images, due to the low segmentation accuracy of the model in the initial stage of model training, the pseudo-label accuracy is also relatively low. In order to reduce the impact of inaccurate pseudo labels, we use the threshold t_u to select pixels with high predicted probability. The loss weight for these selected pixels is set to 1, and others are set to 0. By these thresholds, we filter the supervised loss L_s and the semisupervised loss L_{semi}, so as to reduce model overfitting and make pseudo labels better guide the model training.

In conclusion, the semisupervised method RanPaste mainly mixes the labeled and unlabeled images as model input and then mixes the weakly augmented unlabeled prediction and true labels as pseudo labels. By setting appropriate masks for supervised loss and semisupervised loss, the RanPaste method will continuously learn more latent information from unlabeled remotely sensed images to improve the segmentation performance.

In Section IV, we will demonstrate the advantages and analyze the segmentation performance of the RanPaste method on different remote sensing datasets.

IV. EXPERIMENTS

In order to verify the effectiveness of our RanPaste method on semisupervised remote sensing image segmentation, we conducted experiments on six different remote sensing datasets. We believe that the important significance of our semisupervised method is that RanPaste can improve the segmentation accuracy by using unlabeled images on different types of remote sensing datasets. Thus, we select different types of datasets on our experiments.

As shown in Table I, we compare datasets in our experiment from five aspects, including the number of training samples and test samples, image size, data acquisition methods, spectral channels, and the number of classes. First, for general building and road extraction tasks, we test the effectiveness of our method on INRIA Aerial dataset [38] and road extraction dataset [39]. The difference between them is the amount of training data available and the higher difficulty of the road extraction task. These characteristics are reflected in the experimental results. For the task of multiple classes in semantic segmentation, we choose the iSAID dataset [40], [43] for verification. The dataset contains 15 classes, such as cars and playgrounds. In addition, remote sensing datasets also include remote sensing images collected by UAVs at a relatively low altitude. The AeroScapes aerial semantic segmentation benchmark [41] comprised of images captured using a commercial drone, which contains 11 classes. Finally, some remote sensing datasets contain more spectral channels than conventional RGB images. The 38-Cloud dataset [42] has 8400 patches, and each patch has four corresponding spectral channels that are red, green, blue, and near infrared. Finally, we use the Vaihingen dataset[1] to evaluate our method. It is an airborne image dataset, consisting of very high-resolution treue ortho photo (TOP) tiles and corresponding digital surface models (DSMs) derived from dense image matching techniques. The Vaihingen dataset is divided into six categories. Our method can be directly applied to these datasets, and we also analyze and discuss some additional parameters in our semisupervised method. Implementation details and experimental results will be described in the following.

A. Implementation Details

In our experiments, we used the same three segmented network architectures that are UNet [17], DeepLab V3 [19], and DeepLab V3+ [20]. The purpose of our experiments is to use extensive unlabeled remotely sensed images for the same network to significantly improve the segmentation performance. We used ADAM as our training optimizer during all model training processes and the initial learning rate set as 0.0001. We use a momentum of 0.9 and a weight decay of 0.0005. Our method is implemented with Pytorch.

[1] http://www2.isprs.org/commissions/comm3/wg4/semantic-labeling.html

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All experiments are conducted on a server with an Nvidia Quadro 1080ti graphical processing unit (GPU) (12 GB). Similar to the method of semisupervised classification, the framework of mean teacher is also used as our basic method in the segmentation experiments. During the training process, the update parameter of the teacher model is set to 0.999, and θ_t is updated once every training batch. The regularization coefficient λ is set as 0.1. Our weak image augmentation method η_w corresponds to random image mirroring and flipping, and strong image augmentation η_s applies color jitter and random Gaussian noise.

For the experimental design of the semisupervised method, 1/2, 1/4, and 1/8 of all data are set as labeled data and the rest of the images are selected as unlabeled data for model training on all datasets. We set the batch size as 5, including two labeled images and three unlabeled images, which all guide the training of the model. When we paste the labeled image into the unlabeled image, we select areas where 1/4 the size of the original image is M at a random location. Finally, we choose to use dynamically adaptive values for threshold t_u and t_x. With the training of the model, t_x and t_u gradually increase. We also analyze the change of the threshold in Section V. The selection of these additional parameters will be discussed and analyzed in the subsequent experiments of different datasets. We choose several general semantic segmentation architectures, including UNet, DeepLab V3, and DeepLab V3+. Besides, we also experiment on the latest segmentation network LANet that is designed for remote sensing data. Finally, we conducted our experiments by comparing the proposed semisupervised method with baseline and other methods. To demonstrate the effectiveness and robustness of the proposed RanPaste method, we use mean intersection over union (mIoU) and F1 score as the evaluation metric. For the selection of the training model, we first conducted extensive hyperparameter comparative experiments. After determining the final hyperparameters, we conduct three experiments respectively to test the accuracy of their model and finally used the average value of their results as the final result of the method. We also select some test segmentation results for comparison. For the compared methods, we use default parameters that are specified in their methods.

B. Image Semantic Segmentation on the INRIA Aeria Dataset

INRIA Aeria dataset was constructed by combining public domain imagery and public domain official building footprints. This dataset covers an area of 810 km^2, including densely populated areas and alpine towns. Their ground truth has two semantic classes: building and not building. Our purpose is to train a model to segment the test image accurately and to compare the performance of our semisupervised method. In our experiments, there are 155 training images and 25 test images in this dataset. The remotely sensed images of the training set and the test set to cover five large areas, and different cities are included in each of the subsets. To analyze our semisupervised approach, we set the number of labeled images to 15, 40, and 80. For example, when the number of labeled data is set to 15, the training set has 15 labeled images and 140 unlabeled images. Besides, given the size of the training image is 5000 × 5000, our GPU memory is unable to satisfy requirements. Thus, we crop each image into 500 × 500 small size patches for model training. In order to ensure a fair comparison, we used the same training parameters for different methods of the same network.

TABLE II
SEGMENTATION ACCURACY (mIoU, %) COMPARISON OF RANPASTE ON THE INRIA AERIAL DATASET WHEN DIFFERENT NETWORK STRUCTURES ARE ADOPTED

INRIA Aerial Dataset	15 labels	40 labels	80 labels	155 labels
UNet [17]	67.52	71.50	73.81	75.82
UNet + CRAUP [10]	72.54	73.63	74.95	76.23
UNet + RanPaste	**73.31**	**74.04**	**74.96**	**76.31**
DeepLab V3 [19]	71.97	74.39	75.52	77.61
DeepLab V3 + CRAUP [10]	74.27	**76.21**	**76.92**	78.19
DeepLab V3 + RanPaste	**74.80**	76.12	76.55	**78.23**
DeepLab V3+ [20]	72.30	75.31	76.19	78.35
DeepLab V3+ + CRAUP [10]	74.40	76.64	77.12	78.61
DeepLab V3+ + RanPaste	**75.32**	**76.75**	**77.45**	**78.88**

To test the effectiveness of our semisupervised method and whether it can achieve better performance for different network structures, we train the model with different labeled images of three network structures. After extensive experiments, the experimental results are shown in Table II. From the experiment results, we can learn that when the number of labeled images is reduced, the accuracy of model segmentation will become significantly worse for baseline. Compared with the baseline, our RanPaste improves the model segmentation mIoU from 67.52 to 73.31 by combining more unlabeled remotely sensed images. Compared with the CRAUP method, our method is more simple and efficient. At the same time, the final segmentation performance is better than CRAUP. Through the comparison of three network structures, the results show that our RanPaste method can effectively improve the model accuracy for different network structures. When we have corresponding labels on all the training images, we find that RanPaste also can slightly improve the model segmentation accuracy. From these experiments, we believe that RanPaste can improve the robustness of the segmentation model, and the model will perform better by combining extensive unlabeled images.

C. Image Semantic Segmentation on the Road Extraction Dataset

The satellite imagery used in DeepGlobe for the road extraction challenge is sampled from the DigitalGlobe +Vivid Image dataset. Our task is to accurately extract the road from these images, which can be applied to automated crisis response and increased map coverage for connectivity. The dataset contains 6226 RGB remote sensing images, and the resolution of image data is 1024 × 1024. We divide all data into 5400 training images and 826 test images for our experiments. Similarly, due to the limitation of GPU memory size, we still need to change the training image size. We believe that our purpose is the comparison of semisupervised methods, rather than on achieving the best segmentation performance. Thus, in our all experiments, we resize the image to 512 × 512. We have performed experiments for three proportions of labeled samples,

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TABLE III

SEGMENTATION ACCURACY (mIoU, %) COMPARISON OF RANPASTE ON THE ROAD EXTRACTION DATASET WHEN DIFFERENT NETWORK STRUCTURES ARE ADOPTED

Road Extraction Dataset	400 labels	1350 labels	2700 labels	5400 labels
UNet [17]	48.7	53.55	57.34	59.81
UNet + CRAUP [10]	52.76	**56.12**	58.23	60.02
UNet + RanPaste	**53.20**	55.80	**58.70**	**60.15**
DeepLab V3 [19]	48.42	52.40	54.66	57.23
DeepLab V3 + CRAUP [10]	51.44	**53.68**	55.54	57.67
DeepLab V3 + RanPaste	**51.67**	53.67	**56.57**	**57.74**
DeepLab V3+ [20]	51.50	56.21	59.11	62.41
DeepLab V3+ + CRAUP [10]	53.63	**58.01**	**60.22**	62.63
DeepLab V3+ + RanPaste	**54.24**	57.78	60.19	**62.85**

which are given along with baselines that are trained by a supervised method. For RGB images, our RanPaste method mixes labeled and unlabeled images in the same way, and the weak augmentation and the strong augmentation are controlled by the intensity of image color jitter and random noise.

The experimental results are shown in Table III. For the experiments of baseline, when we performed supervised training with 5400 labels, the mIoU of the segmentation model is 59.81. However, when the number of labeled images decreased to only 400, the performance of the model decreased significantly. In UNet experiments of different semisupervised approaches, CRAUP can improve the model's mIoU to 52.76 by using only 400 labeled images, while our RanPaste method is able to further improve the model's performance to 53.20. These results show that the semisupervised segmentation method can effectively use unlabeled image latent information to improve the original model. However, it is also worth noting that due to the relatively small target of the road extraction task, our RanPaste method may not achieve better results for the boundary areas when different images are mixed. Compared with experiments of different network structures, our RanPaste can achieve better segmentation performance for semisupervised remote sensing image semantic segmentation.

D. Image Semantic Segmentation on the iSAID Dataset

In the image semantic segmentation task, multiclass image segmentation is an essential part of many applications. For remote sensing data, it is also important research to segment objects in remote sensing images and classifies them. Given these reasons, we also test the proposed semisupervised method on the iSAID dataset. The data of iSAID are mainly collected from Google Earth, satellite JL-1, and GF-2. This dataset contains 15 classes to be segmented, such as plane, bridge, and swimming pool. In our experiment, 1200 RGB remote sensing images are used as the training set, and 211 images are used as the test set. Since the size of the images contained in this dataset is different, we resize all images to 512×512 in the process of the training model and testing model. Our semisupervised experiment used the same set of image numbers as CRAUP experiments, while other experimental parameters remained unchanged.

We use the same networks and hyperparameters as CRAUP and keep the same number of unlabeled images and labeled images to ensure that the semisupervised methods can be compared fairly. We conduct semisupervised experiments on the iSAID dataset, and numerical results are shown in Table IV. When the number of labeled images was reduced, the mIoU for the standard supervised training method on the UNet network dropped from 36.92 to 25.13. The result illustrates the important impact of the amount of labeled data on DL methods, and our research goal is to alleviate this problem by using unlabeled data. Add the CRAUP method to the basic method, and when there are only 100 labeled images, the mIoU of the model increases from 25.13 to 29.51. Compared with the CRAUP method, our RanPaste further improves the model performance, and the final mIoU is 33.51. Our semisupervised method RanPaste only used 100 labeled remote sensing images and more unlabeled images, and the final model performance reached the model results that trained in 600 labeled images. Experimental results on three different network structures show that our RanPaste can effectively improve the model segmentation performance. In addition, compared with CRAUP, our method is more obvious in improving the performance, and the result is better than previous methods.

TABLE IV

SEGMENTATION ACCURACY (mIoU, %) COMPARISON OF RANPASTE ON THE iSAID DATASET WHEN DIFFERENT NETWORK STRUCTURES ARE ADOPTED

iSAID Dataset	100 labels	300 labels	600 labels	1200 labels
UNet [17]	25.13	27.59	33.50	36.92
UNet + CRAUP [10]	29.51	31.12	35.17	37.24
UNet + RanPaste	**33.51**	**34.26**	**35.76**	**37.37**
DeepLab V3 [19]	36.13	39.12	41.85	44.86
DeepLab V3 + CRAUP [10]	38.92	40.82	42.95	45.13
DeepLab V3 + RanPaste	**40.34**	**41.88**	**43.40**	**45.24**
DeepLab V3+ [20]	36.76	41.46	43.92	47.45
DeepLab V3+ + CRAUP [10]	39.85	42.88	45.02	47.73
DeepLab V3+ + RanPaste	**41.45**	**43.15**	**45.65**	**47.88**

In order to directly compare the segmentation performance of different methods, we show some segmentation results of the model on the test set on DeepLab V3+. The model segmentation results of different methods are shown in Fig. 4. As can be seen from the figure, when only 100 remote sensing images with labels are used to train the model, the segmentation results of the model become rough, and many targets that need to be segmented are lost. To make a fair comparison between RanPaste and CRAUP, we choose two same test images for comparison. CRAUP's prediction results in Fig. 4 are the same as their paper, and our RanPaste results are the segmentation results obtained on the same network structure. It is obvious from the figure that our RanPaste method can obtain better segmentation results, which also indicates that our semisupervised method is more effective. Compared with different semisupervised remote sensing image segmentation methods, the segmentation results obtained by RanPaste are closer to the ground truth. From the results of these models, our RanPaste method can effectively utilize unlabeled images to improve the segmentation performance of the segmentation model.

E. Image Semantic Segmentation on the Aeroscapes Dataset

The datasets of our previous experiments are all collected by satellite or aerial photography. For remote sensing images,

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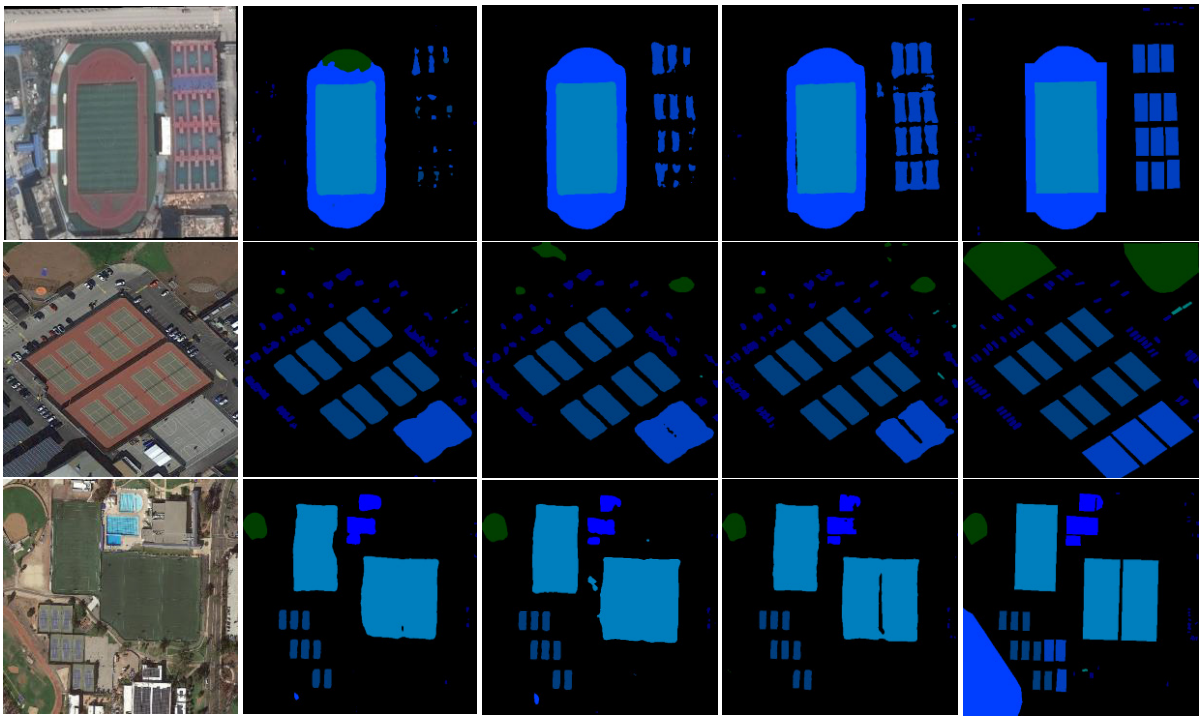


Fig. 4. Some examples of remote sensing image segmentation when different methods are adopted on validation set on the iSAID dataset. (a) Input. (b) 100 labels. (c) CR + AUP. (d) RanPaste. (e) Ground truth.

images collected by UAV are also an important part of remote sensing data. To verify the performance of our semisupervised segmentation method on different types of remote sensing images, we selected Aeroscapes as the dataset of our experiments. The Aeroscapes aerial dataset comprises images captured using a commercial drone from an altitude range of 5–50 m. This dataset contains a total of 3269 RGB images with a size of 1280 × 720, and 11 categories need to be segmented. In our experiments, we resize all images to the size of 640 × 360 and divide them into 2400 training sets and 869 test sets. For semisupervised experiments, we divide all the training images into several quantities according to the previous similar ratio to test our semisupervised method.

We first conducted basic supervised experiments on different network structures to record the mIoU of models with different amounts of labeled data. The experimental results show that the number of available labeled images will greatly affect the performance of the model. Both DeepLab V3 and DeepLab V3+ decrease in mIoU values of 2%–3% due to the decrease in labeled images. We can see from Table V that when we used RanPaste to train the model, the mIoU of the UNet model increased from 23.83 to 30.01 by using only 300 labeled images. Experiments on different amounts of labeled data show that the RanPaste method can significantly improve the segmentation performance of the model. However, it is worth noting that when we conducted semisupervised experiments on DeepLab V3 and DeepLab V3+ networks, we find that RanPaste does not greatly improve the model performance. We believe that the main reason is that they are less dependent on the amount of labeled data on these architectures. Therefore, our semisupervised method has a limited improvement on the model.

TABLE V
SEGMENTATION ACCURACY (mIoU, %) COMPARISON OF RANPASTE ON THE AEROSCAPES DATASET WHEN DIFFERENT NETWORK STRUCTURES ARE ADOPTED

Aeroscapes Dataset	300 labels	600 labels	1200 labels	2400 labels
UNet [17]	23.83	26.23	31.22	33.93
UNet + RanPaste	**30.01**	**31.38**	**32.51**	**34.21**
DeepLab V3 [19]	38.36	39.88	40.47	41.12
DeepLab V3 + RanPaste	**39.68**	**40.66**	**41.18**	**42.16**
DeepLab V3+ [20]	39.08	40.14	40.86	41.71
DeepLab V3+ + RanPaste	**39.71**	**41.36**	**41.92**	**42.20**

We also show some segmentation results of different methods on the test set. As shown in Fig. 5, the original image mainly includes roads, trees, persons, and other targets that need to be segmented. The original model was able to segment the target when there was less labeled training data, but it misidentified the vehicle as a person. It is also clear from the figure, which shows the severe impact of reducing labeled data on the model performance. For the segmentation results of the third column, our RanPaste obtains more accurate segmentation predictions. As can be seen from the figure, our method changes the misidentified class into the correct category, and the segmentation areas are more accurate. Combining the mIoU of different models and their segmentation predictions, it can be seen that RanPaste can make full use of the underlying information contained in unlabeled data to enhance the model performance.

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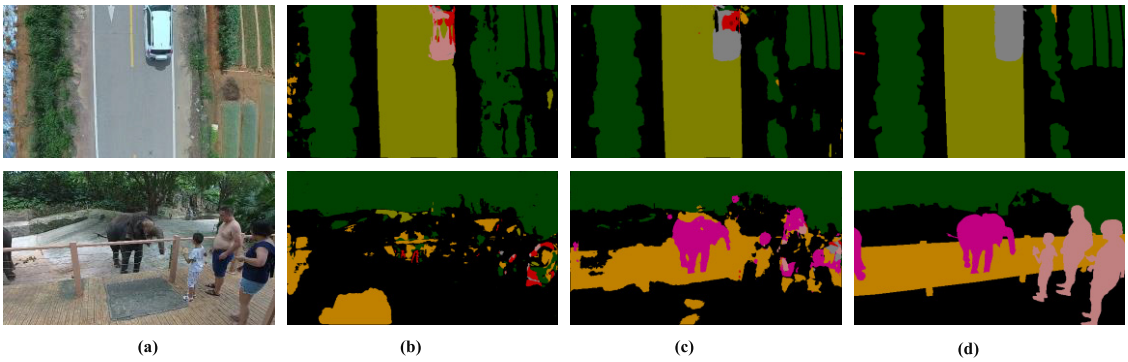


Fig. 5. Segmentation maps produced by RanPaste on the aeroscapes dataset. (a) Input. (b) 300 labels. (c) RanPaste. (d) Ground truth.

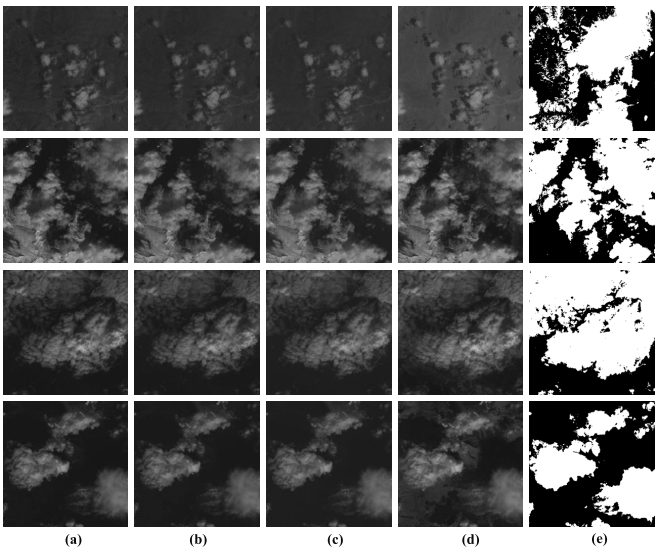


Fig. 6. 38-Cloud dataset. (From Left to Right) Four channels of remote sensing data and the ground truth. (a) Red. (b) Green. (c) Blue. (d) Nir. (e) Ground truth.

F. Image Semantic Segmentation on the 38-Cloud Dataset

In order to test the effectiveness of our semisupervised method on different data types of remote sensing data, we also conduct experiments on multispectral remote sensing images. The 38-Cloud dataset contains 38 Landsat 38 scene images, and the dataset also provides pixel-level ground truths for cloud detection. Since the original scenes contain too many areas without information, we randomly cropped the data into 384 × 384 patches. Unlike general computer vision images, these channels are not combined. As shown in Fig. 6(a)–(d), each patch has four corresponding spectral channels that are red, green, blue, and near infrared. Besides, Fig. 6(e) are their pixel-level ground truths. Finally, removing most of the blank patches, we use 5400 patches as the training set and 955 patches as the test set. Our semisupervised experiment sets different numbers of labeled data that are 525, 1050, and 2100. For different network structures in the experiment, we change input channels to 4. Other settings are consistent with the other group's experiments.

The main comparative experiment was to train supervised models on three network structures as baseline and then train segmented models by using our RanPaste semisupervised approach. Given that the network input channels in this experiment need to be modified into four channels, DeepLab's networks cannot load the pretrained ResNet-50 model parameters at the beginning. The results of our comparative experiments are shown in Table VI. It can be seen from the table that when the pretrained model cannot be loaded, the mIoU of UNet models is closer to the results of DeepLab's networks than other experiments. From the experimental results, we learn that when the labeled training data are reduced, combining unlabeled data and RanPaste's model, mIoU on three networks has increased 2–3 percentage points. These results show that RanPaste can also perform well in

TABLE VI
SEGMENTATION ACCURACY (mIoU, %) COMPARISON OF RANPASTE ON THE 38-CLOUD DATASET WHEN DIFFERENT NETWORK STRUCTURES ARE ADOPTED

38-Cloud Dataset	525 labels	1050 labels	2100 labels	4200 labels
UNet [17]	64.34	66.51	68.88	70.35
UNet + RanPaste	**66.57**	**68.59**	**70.20**	**70.55**
DeepLab V3 [19]	63.91	65.64	67.50	69.25
DeepLab V3 + RanPaste	**66.96**	**67.56**	**68.82**	**69.54**
DeepLab V3+ [20]	65.80	67.68	68.70	69.95
DeepLab V3+ + RanPaste	**67.82**	**68.94**	**69.36**	**70.03**

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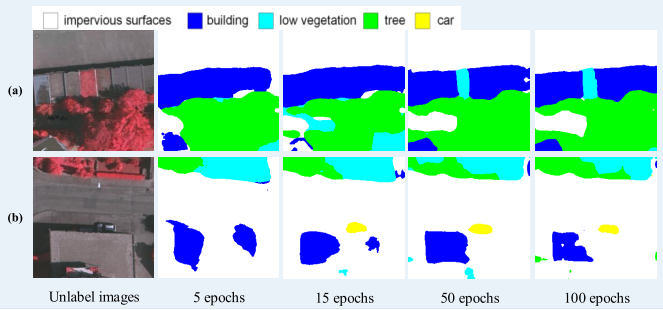


Fig. 7. Some examples of pseudo labels for unlabeled image. Pseudo label updates with training epochs on the Vaihingen dataset. (a) Unlabeled sample a. (b) Unlabeled sample b.

semisupervised segmentation of multispectral remote sensing data.

G. Image Semantic Segmentation on the Vaihingen Dataset

To verify the applicability of our semisupervised method for multiclass semantic segmentation of remote sensing images, we also conduct a large number of experiments on the Vaihingen dataset. Besides, we select the latest RS-oriented segmentation algorithm LANet [24] for comparison. LANet is an improved architecture for remote sensing image segmentation. In these experiments, we use UNet and LANet as our baseline to compare with our method.

The Vaihingen dataset contains 16 TOP tiles and the corresponding DSMs collected from a small village. In our experiments, we only use the TOP images as our training data. The spatial resolution of this dataset is 9 cm, and the spatial size of the images varies from 1996 $\times$ 1995 pixels to 3816 $\times$ 2550 pixels. We select 12 original images as the training set and four images as the test set. Due to the limitation of computational resources, we use a 256 $\times$ 256 sliding window to crop the training samples from the original images. By this way, we finally obtain 760 training images and 280 test images for our experiments. For the semisupervised experiment, we select 1/8 of the training set as our labeled data, so we have 100 labeled images and 660 unlabeled images for training.

In the training process, we also evaluate the quality of pseudo labels. As shown in Fig. 7, we show pseudo labels for some unlabeled samples. We use models of different epochs to generate the corresponding pseudo labels in the training process. As we can see from the figure, the pseudo-label quality of the model gradually improves with the training process. Because of the threshold t_u, more pixels in the unlabeled images are added to the semisupervised loss so that the segmentation model can be further improved by using the unlabeled images.

We report the performance of all baselines along with RanPaste in Table VII. We first conduct fully supervised experiments with only 1/8 labeled images on UNet and LANet and record the F1 scores of the five categories and their mIoU. From the results, we find that LANet has a better performance overall. In order to compare our semisupervised method, we use RanPaste to train the semisupervised segmentation model combining the remaining unlabeled images. Experimental results show that our semisupervised approach

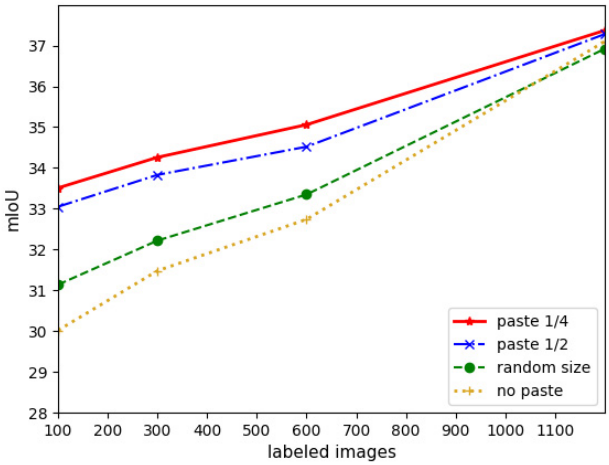


Fig. 8. Comparison of semisupervised results obtained using different paste sizes on the iSAID dataset.

achieves good performance on different networks by better leverage unlabeled data. The semisupervised model can improve the average F1 by about six percentage points. For the latest remote sensing image segmentation model LANet, the mIoU of the semisupervised model also increases from 57.91 to 65.56. All these results prove that our RanPaste method can effectively improve the segmentation model by using unlabeled data.

V. Discussion

After extensive experiments, results show that RanPaste achieves our goal of improving semisupervised semantic segmentation. Besides, there are some details in our approach that are worth analyzing.

A. Ablation Study

First, our RanPaste pastes the labeled image into unlabeled image with a mask M. There are many methods to select M, we have also conducted sufficient experimental analysis. We test the effects of different pasted areas by using RanPaste on the iSAID dataset, and the basic architecture is UNet. The experimental results are shown in Fig. 8. We first conducted a set of experiments using only unlabeled images instead of mixing them, which are no paste in Fig. 8. Then, we experimented with three types for M: random size, 1/4 of original size, and 1/2 of original size. From these experiments, we find that the performance of the model is significantly improved after the combination of labeled and unlabeled images. The RanPaste method obtains the best results when the size of M is 1/4 of the original image size. We believe that when the area occupied by labeled images is too large, it will affect the training of the semisupervised model, thus leading to worse results. In our experiments, we always keep M to 1/4 of the size of the original image.

As for the different thresholds used in loss functions, we also design experiments to select the most appropriate parameters. First, we do not change the loss with the supervised training, and then, we make an experimental comparison of the threshold values in pseudo-label generation. These semisupervised experiments were also conducted using UNet on the

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TABLE VII

SEGMENTATION MIOU (%) AND F1 SCORES (%) COMPARISON OF RANPASTE ON THE VAIHINGEN DATASET WHEN DIFFERENT NETWORK STRUCTURES ARE ADOPTED

Method with 1/8 labels	Per-class F1 Score (%)					Average F1 (%)	MIoU (%)
	Impervious Surface	Building	low vegetation	Tree	Car		
UNet [17]	79.71	84.56	60.03	80.74	54.12	71.83	57.44
UNet + RanPaste	**82.43**	**86.27**	**67.30**	**83.28**	**65.71**	**77.00**	**63.39**
LANet [24]	79.95	85.03	61.53	81.53	52.67	72.14	57.91
LANet + RanPaste	**83.39**	**87.88**	**69.03**	**83.28**	**65.65**	**77.84**	**64.56**

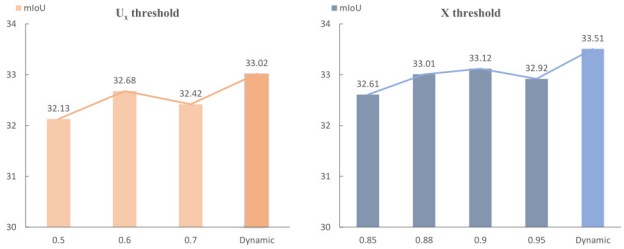


Fig. 9. Selection of loss thresholds t_x and t_u. (Left) Threshold t_u when unlabeled data generate pseudo labels. (Right) Threshold t_x when supervised loss is calculated.

iSAID dataset. The threshold value of the prediction result of the usual segmentation model is set to 0.5. However, as shown in Fig. 9, when we slightly increase the threshold value of pseudo-label generation in the semisupervised training, the model performance can be clearly improved. Then, as can be seen from the figure, dynamically changing thresholds can significantly improve the performance of the model, and there is no need to manually set thresholds for different datasets. Finally, we set the threshold t_u to a dynamic value for generating pseudo labels on unlabeled images. In order to prevent overfitting of labeled data, we also set a threshold for supervised loss. Our comparative experiment is to train the semisupervised model by constantly changing the threshold of labeled images under the condition that the pseudo label threshold t_u is a dynamic value. This threshold value is 1 by default in general supervised training. However, in the case of less labeled data in the semisupervised experiment, we reduce the threshold value to effectively alleviate the overfitting problem. After experiments of these groups, we finally determine that the thresholds t_x and t_u for the training of labeled images and unlabeled images are values that change dynamically with the performance of the model.

In addition, we also analyze the change of threshold value during the model training process. When we used UNet to conduct semisupervised experiments on the Aeroscapes dataset, the change curves of threshold t_x and t_u are shown in Fig. 10. Since the threshold we used was directly related to the predicted performance of the model, it is also obvious from the figure that the threshold was a relatively small value at the beginning of the experiment. With the improvement of model accuracy, our thresholds gradually become large, so the pixel participating in the training also requires greater prediction probability These methods also achieve the goal of setting thresholds. The final experimental results also show

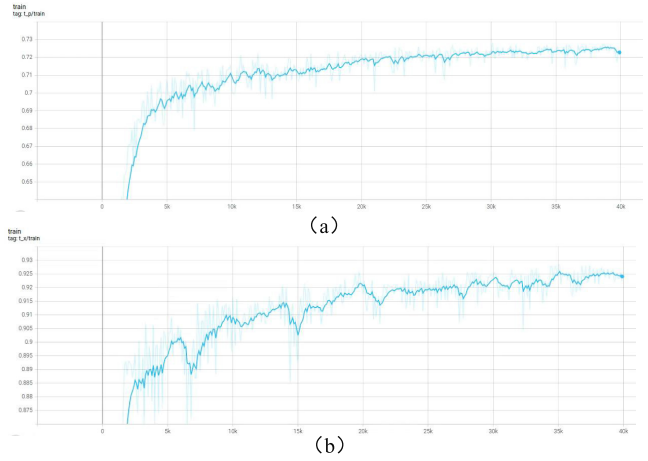


Fig. 10. Comparison of the behaviors of training thresholds through the training process on the Aeroscapes dataset. (a) Pseudo label generation threshold t_u versus iterations. (b) Supervised training threshold t_x versus iterations.

TABLE VIII

HYPERPARAMETER ANALYSIS AND ABLATION EXPERIMENTS

Method	t_u	t_x	Paste Size	MIoU (%)
UNet [17]	-	-	-	25.13
UNet + t_u	0.5	-	-	26.04
	0.6	-	-	26.87
	t_u_d	-	-	**27.12**
UNet + t_u + t_x	t_u_d	0.85	-	27.04
	t_u_d	0.9	-	28.22
	t_u_d	t_x_d	-	**28.43**
UNet + t_u + t_x + Paste	t_u_d	t_x_d	Random Size	31.24
	t_u_d	t_x_d	1/4	**33.51**
	t_u_d	t_x_d	1/2	33.02

that dynamic thresholds are better than the fixed thresholds set manually.

We also conduct experiments with different hyperparameters to analyze our method. We choose iSAID as our experimental dataset and use UNet with 100 labels as a baseline. The results of the experiment are shown in Table VIII. First, we only combine t_u with pseudo labels in semisupervised experiments. We record the segmentation performance of the model by adjusting the size of t_u. Combining the model accuracy of fixed t_u and dynamic t_u_d, the experimental results show that the adaptive threshold has a better performance. Then, we fix t_u_d and add t_x for the labeled data loss. After comparing several parameters, we find that the adaptive threshold t_u_d also has a better result. Finally, to analyze the influence of

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TABLE IX
TIME COMPLEXITY ANALYSIS OF RANPASTE

Architecture	Method	Train time	Params	MIoU (%)
UNet on iSAID dataset	UNet [17]	**1.53 h**	17.2 M	25.13
	UNet + RanPaste	9.14 h	**17.2 M**	**33.51**
LANet on Vaihingen dataset	LANet [24]	**0.11 h**	23.8 M	57.91
	LANet + RanPaste	0.70 h	**23.8 M**	**64.56**

different RanPaste sizes, we keep the two threshold methods and conduct a group of comparative experiments with different paste sizes. The experimental results are shown in the table. After comparing the random size, 1/2 original image size, and 1/4 original image size, the semisupervised method has the best performance when the RanPaste size is 1/4 image size. Through the comparison experiments of these hyperparameters, we determined the parameter selection method of RanPaste method in the experiment. Combining these methods, our semisupervised method obtains satisfactory results.

B. Time Complexity

Although our semisupervised method has achieved good performance, we also need to analyze the time complexity of RanPaste. All of our comparative experiments were conducted on a 1080ti GPU with the same experimental environment, so as to make a fair comparison of the complexity of the methods. As shown in Table IX, we use two architectures to conduct experiments on different datasets. First, a fully supervised model of 100 labels images was trained on the iSAID dataset by UNet, and the number of model parameters and the time spent in training were recorded. It can be seen that the model has 17.2M parameters, and the training time is only 1.53 h, but at the same time, the corresponding segmentation results also have a low mIoU value. When we used the RanPaste method to train the semisupervised model, the number of model parameters did not change. Because our method mainly improved the training loss and pseudo-label method in the training process, we keep the original model size. It should also be noted that due to the complexity of training model, the training process of our model is increased to 9.14 h. More importantly, the model mIoU was significantly improved by the RanPaste method.

In addition, we have also used the latest designed LANet for remote sensing images for semisupervised experiments. Through experimental comparison, the RanPaste method also increases the training time of the model from 0.11 to 0.7 h but improves the model performance at the same time. Finally, although our method improves the training time of original model, we do not add any additional time consumption in the test phrase because model parameters remained unchanged. Combined with these experiments, we believe that RanPaste can obtain a more accurate segmentation model at the cost of increasing the training time. This also achieves the goal of semisupervised segmentation of remote sensing images.

We conducted our semisupervised experiments on six different types of remote sensing image datasets. The experimental

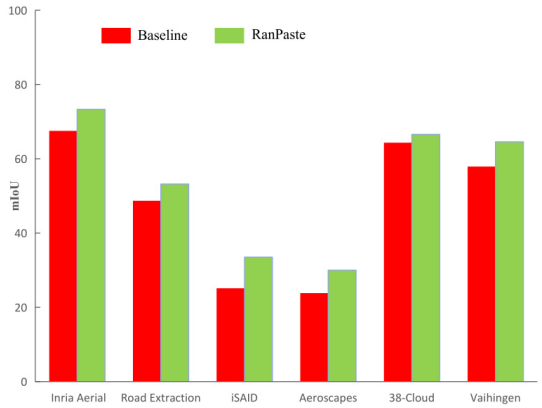


Fig. 11. Comparison of RanPaste's performance on five remote sensing datasets.

results show that our RanPaste method can improve the segmentation model by utilizing information from unlabeled images. We combined RanPaste's results on the six datasets, as shown in Fig. 11. It can be seen from the figure that our proposed semisupervised approach has a better performance than the baseline on six datasets. Besides, we also notice that our semisupervised approach has more significantly improved for multiclass semantic segmentation tasks. There are some existing semisupervised segmentation methods, which mainly generate pseudo labels by GANs networks. Compared with them, the pseudo labels generated by our RanPaste model are closer to the real data distribution. For other consistence semisupervised methods, they direct mix unlabeled images for training. Our pasting method can better combine the information of labeled images and unlabeled images. It can obtain the target edge features better. Our experiments also prove that these differences make our semisupervised method performs better.

In general, compared with other semisupervised remote sensing image segmentation methods, the RanPaste method combines the consistency method and directly integrates pseudo labels into the model training process. Through the design of threshold values, we do not need to adjust the weight of semisupervised loss, because through the improvement of model accuracy, more information contained in unlabeled data will be learned by the model. Of course, our semisupervised approach also has some shortcomings. On the multispectral dataset, RanPaste only has limited improvement on the segmentation model. We think that our image pasting method maybe is more suitable for general natural images. In future works, we will explore more effective methods for multispectral remote sensing datasets. Besides, because of the

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threshold, our RanPaste needs to train more epochs during the semisupervised training process. However, for the test phase, our method has the same inference speed as others methods. We believe that these problems are acceptable for a semisupervised approach.

VI. Conclusion

Remote sensing image semantic segmentation usually requires a large number of pixel-level annotated images, but the annotation of remote sensing images is expensive in practical applications. In this article, we propose a semisupervised semantic segmentation method RanPaste for remote sensing images. This method can effectively use extensive unlabeled remote images to improve the performance of the segmentation model. Compared with other semisupervised methods, RanPaste can effectively combine labeled images and unlabeled images, and the model will learn more information from unlabeled images. After experiments on six different types of remote sensing datasets, results show that RanPaste can significantly improve the segmentation performance by using unlabeled images. At the same time, our approach can achieve the segmentation task on different network structures and different datasets.

In the future, we will try to further improve the semisupervised algorithm for remote sensing image binary segmentation. Besides, we will explore how to better combine the information of unlabeled data to improve the segmentation model when there are only few labeled images. We believe that semisupervised learning will play a greater role in the practical application of remote sensing images in the future.

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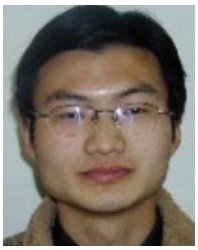
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